

SF-EDGE Incident and Change Log

Document ID: SF-EDGE-OPS-005

Version: 0.9

Date: 2026-05-09

Owner: Operations Readiness

Project: Smart Factory Edge Gateway

1. Purpose

This document records selected incidents, test findings, and change decisions observed during pre-release validation of SF-EDGE.

The objective is to provide operational context for future test planning and knowledge base queries.

2. Incident INC-2026-041: Duplicate Telemetry After Cloud Timeout

Date detected: 2026-04-29

Severity: High

Status: Fixed

Related components:

- Local Buffer;
- Cloud Forwarder;
- cloud ingestion simulator.

Description

During a simulated cloud timeout, the Cloud Forwarder retried a batch after the cloud simulator had already accepted part of the batch.

The simulator detected duplicate event IDs for 37 telemetry records.

Root Cause

The forwarder marked the full batch as failed when the HTTP response timed out. It did not reconcile partial acceptance using event IDs.

Fix

The forwarder now sends idempotency keys and reconciles accepted event IDs after timeout recovery.

Regression Tests Required

- replay buffered telemetry after timeout;
- verify no duplicate records are accepted;
- verify `retry_count` is updated only for records still pending;
- verify audit log records the forwarding anomaly.

Related requirements:

- REQ-FUNC-005;
- REQ-NFR-005.

3. Incident INC-2026-047: Diagnostic Bundle Exposed Token-Like Value

Date detected: 2026-05-01

Severity: Critical

Status: Fixed

Related components:

- Diagnostic Bundle;
- Redaction Utility;
- Audit Service.

Description

A diagnostic bundle contained a token-like value in a nested JSON field named `cloud_debug_header`.

Root Cause

The redaction utility scanned only known top-level keys. It did not recursively inspect nested JSON fields.

Fix

Redaction now recursively scans strings and keys using secret pattern detection.

Regression Tests Required

- nested JSON secret redaction;
- private key redaction;
- bearer token redaction;
- diagnostic export audit event;
- checksum recording for exported bundle.

Related requirements:

- REQ-FUNC-010;
- SEC-REQ-021;
- SEC-REQ-052.

4. Incident INC-2026-052: Device Health Update Delay

Date detected: 2026-05-04

Severity: High

Status: Open

Related components:

- Protocol Adapter Runtime;
- Device Health Service.

Description

During an OPC UA disconnect test, the device health status changed to disconnected after 47 seconds instead of the expected 30 seconds.

Suspected Cause

The OPC UA adapter waits for two full session timeout cycles before reporting failure.

Temporary Mitigation

The health service now displays degraded status after the first failed communication cycle.

Tests Required

- OPC UA disconnect detection;
- Modbus disconnect detection;
- MQTT broker disconnect detection;
- HTTP sensor timeout detection.

Related requirements:

- REQ-FUNC-006;
- REQ-FUNC-007.

5. Change Decision CHG-2026-018: Buffer Retention

Date: 2026-05-05

Decision owner: Product Engineering

Status: Approved for version 1

Decision

The version 1 buffer retention requirement remains 24 hours at 2,000 measurements per minute.

A 72-hour buffer target is deferred to a later release.

Rationale

Reference hardware storage is limited to 64 GB. The 24-hour target is sufficient for the initial customer pilot.

Related requirement:

- REQ-FUNC-004.

6. Change Decision CHG-2026-019: Automatic Certificate Rotation

Date: 2026-05-06

Decision owner: Cybersecurity Team

Status: Deferred

Decision

Automatic OPC UA certificate rotation is not included in version 1.

The gateway shall still alert administrators 14 days before certificate expiration.

Related requirement:

- SEC-REQ-022.

7. Operational Notes

Known high-value test areas:

- local buffering during cloud outage;
- duplicate prevention after transient failures;
- support grant expiration;
- diagnostic redaction;
- health state transition timing;
- audit completeness after configuration changes.

8. Useful Knowledge Base Questions

The following questions should return meaningful answers if the knowledge base is working correctly:

1. Which requirements are related to duplicate telemetry prevention?
2. What happened in incident INC-2026-047?
3. Which tests should be run for diagnostic bundle redaction?
4. Is 72-hour local buffering required in version 1?
5. Which roles can register a device?

6. What should happen if cloud TLS validation fails?
7. Which open issue affects device health timing?